The periodic table (1)

Elements such as carbon, gold and copper have been known since ancient times. Hydrogen was discovered in 1766. By 1800, scientists had started sorting the elements into groups. At that time many elements had not yet been discovered. In 1869, a Russian scientist named Dimitri Mendeleev ordered the elements according to how heavy they were. He displayed this information in a chart and called it the **periodic table.**

Look at the periodic table on the inside front cover of this book. The periodic table contains more than 100 elements, but only 92 of them are found naturally on earth. The rest are man-made. Elements that have similar properties are grouped together in columns in the table. For example, helium (He) has similar properties to neon (Ne) – both are gases that do not react – so they are grouped together.

The elements grouped on the left-hand side of the periodic table are **metals**. The **non-metals** are grouped on the right-hand side.

Each element has a **chemical symbol**. Some symbols have one letter, while others have two or three. For example, C is the symbol for carbon. Chlorine, which was discovered after carbon, has the symbol Cl. Some symbols come from the element's Latin name. The symbol for iron is Fe, which comes from its Latin name, *Ferrum*.

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The periodic table (2)

☞ Use the Resource sheet 'The periodic table (1)' and the periodic table on the inside front cover to answer the following questions.

1 What is listed in the periodic table? _____

2 Who first developed the periodic table? _____

3 In what year was the periodic table first developed? _____

4 Find out the symbol for the following elements. If the symbol comes from the Latin name, write the symbol in red pen.

a magnesium	_____	**e** sodium	_____	**i** carbon	_____
b oxygen	_____	**f** aluminium	_____	**j** gold	_____
c nitrogen	_____	**g** iodine	_____	**k** potassium	_____
d sulphur	_____	**h** zinc	_____	**l** chlorine	_____

5 Which of the elements listed in Question 4 are non-metals?

_____

_____

6 Sort the substances below into the correct groups.

water iron sand air carbon aluminium silver oxygen carbon dioxide helium plastic sulphur calcium steel sodium salt argon

Element	Not an element

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